

Enceladus: Everything you need to know about Saturn's bright, icy moon

<https://www.space.com/20543-enceladus-saturn-s-tiny-shiny-moon.html>

Enceladus is an intriguing small moon orbiting Saturn, classified as the sixth-largest of Saturn's many moons, and is considered one of the most promising locations in our solar system for the potential existence of life. It was discovered by the British astronomer Sir William Herschel on August 28, 1789, using the world's largest telescope at the time. Remarkably, he also discovered the moon Mimas shortly after, showcasing his contributions to lunar astronomy.

The surface of Enceladus is strikingly bright and composed primarily of clean, reflective ice, making it the brightest object in our solar system. However, its surface is not uniform; it features a combination of heavily cratered regions along with smoother areas that exhibit little to no cratering, suggesting ongoing geological activity. This mix indicates that Enceladus has undergone recent geological processes that allow it to renew its surface.

One of the defining characteristics of Enceladus is its geysers, which continuously eject a mixture of water vapor, carbon dioxide, methane, and other potential life-sustaining chemicals into space at speeds of approximately 800 miles per hour. These plumes extend hundreds of miles into space and contribute material to Saturn's E-ring. While some of this material falls back onto the moon's surface, a significant portion escapes into orbit around Saturn. The interaction of the ejected material with sunlight and Saturn's magnetosphere is an area of active scientific research.

Additionally, it is hypothesized that Enceladus harbors a liquid subsurface ocean, kept in a liquid state due to tidal heating caused by the moon's varying proximity to Saturn. The gravitational forces exerted by the gas giant led to intense tidal flexing, generating friction and heat within Enceladus' interior.

Recent studies have suggested that this subsurface ocean could contain all the essential chemical ingredients necessary for life. The Voyager spacecrafts, specifically Voyager 1 and Voyager 2, were among the first to gather significant data on Enceladus, followed by the Cassini spacecraft, which provided in-depth photographic and chemical analyses of the moon.

In terms of potential habitability, the plumes that erupt from Enceladus are believed to contain salts and tiny grains of silica, which on Earth form in the presence of hot water interacting with rock, indicating the potential for hydrothermal vents on the ocean floor. This is particularly compelling because such environments are thought to be conducive to the origins of life on Earth.

Future missions to Enceladus are proposed, aiming to investigate its geological features, chemical makeup, and the feasibility of sustaining life. Scientists are particularly curious about the possibility that microorganisms could be present in the expelled water from the geysers, further emphasizing the significance of this icy moon in the quest to understand life beyond Earth.

Your Guide to Water on the Moon

<https://www.planetary.org/articles/water-on-the-moon-guide>

The Moon features permanently shadowed regions (PSRs) at its poles that have been found to contain substantial amounts of water ice. These intriguing areas are of great interest to space agencies, particularly NASA, which is planning missions like VIPER to thoroughly explore the lunar poles and their potential resources.

The existence of water on the Moon has intrigued scientists for decades. While researchers long suspected that water might be present, definitive evidence was not obtained until 2008. This milestone discovery occurred when India's space agency ISRO launched the Chandrayaan-1 spacecraft, which included a specialized instrument called the Moon Mineralogical Mapper (M3). This instrument was crucial in analyzing how the lunar surface absorbed infrared light and revealed the presence of water ice in the Moon's polar craters.

A year later, in 2009, NASA further confirmed the presence of water through the Lunar Reconnaissance Orbiter (LRO) and a dedicated impact probe known as LCROSS. The upper stage of the rocket that launched LRO and LCROSS was intentionally crashed into a dark crater at the Moon's south pole. As LCROSS traversed the resulting debris plume, it detected about 155 kilograms of water before colliding with the Moon itself.

The origins of this lunar water are believed to be tied to historical events in our solar system. Water-bearing comets and asteroids have impacted Earth and the Moon multiple times, depositing water upon their surfaces. Additional theories suggest that some of the Moon's water could have been sourced from Earth, especially if the popular hypothesis regarding the Moon's formation—stemming from a colossal impact between the early Earth and another celestial body—holds true. While most of the Moon's water may have evaporated into space over the eons, some managed to become trapped in the PSRs, where it has remained preserved for billions of years despite the harsh lunar environment.

Interestingly, even in non-polar regions, solar activity plays a role in producing water on the Moon. Protons emitted by the Sun interact with oxygen molecules in the lunar soil, leading to the creation of water, though in extremely small amounts. Consequently, the lunar regolith remains significantly drier than Earth's arid deserts.

Current estimates suggest that the lunar poles harbor over 600 billion kilograms of water ice, which is enough to fill at least 240,000 Olympic-sized swimming pools. This estimate is considered conservative, as it is constrained by the limits of the radar technology used by current spacecraft. Future missions equipped with more advanced radars that can penetrate the surface more effectively are anticipated to discover additional deposits of water ice. Notably, in 2020, research utilizing data from LRO identified PSRs within smaller craters, further increasing the estimated total of water ice on the Moon by 10 to 20 percent. These findings position the Moon as a potential resource for future astronauts, who could utilize the ice for breathing air, drinking water, and creating propellant for space travel.

Europa Clipper: A complete guide to NASA's astrobiology mission

<https://www.space.com/europa-clipper-mission-explained>

NASA's Europa Clipper mission is an ambitious endeavor designed to investigate Jupiter's intriguing moon Europa, particularly focusing on its subsurface ocean in an effort to evaluate its potential for supporting life. Launched on October 14, 2024, from NASA's Kennedy Space Center in Florida aboard a SpaceX Falcon Heavy rocket, the spacecraft is set to embark on a journey that will span approximately five and a half years, culminating in its arrival in the Jovian system on April 11, 2030.

Upon reaching Jupiter, Europa Clipper will establish a long, elliptical orbit around the gas giant, which allows it to minimize exposure to Jupiter's intense radiation environment. This careful orbital design is crucial, as Jupiter's immense magnetic field creates a magnetosphere filled with high-energy charged particles capable of damaging spacecraft electronics. To mitigate this risk, Europa Clipper will primarily orbit far from Jupiter, periodically swooping in for close encounters with Europa to conduct in-depth scientific investigations.

During its mission, the spacecraft is scheduled to make nearly 50 close flybys of Europa. Each flyby will provide valuable data, as it passes over various regions of the moon's surface. Scientists have long speculated that beneath Europa's thick ice shell, which ranges from 2 to 20 miles (3 to 30 kilometers) in thickness, lies a vast, saltwater ocean. Previous exploration efforts by missions such as Pioneer, Voyager, Galileo, Juno, and others have gathered compelling evidence of this ocean's existence.

The Europa Clipper's primary objective is to assess whether the conditions within this subsurface ocean might be conducive to life. Equipped with state-of-the-art scientific instruments, the spacecraft will analyze the moon's surface and subsurface properties, seeking to understand its chemistry, geology, and potential habitability. The findings from the Europa Clipper mission could significantly advance our understanding of the potential for life beyond Earth and elucidate the conditions necessary for life to thrive in extraterrestrial environments.

Europa: A guide to Jupiter's icy moon

<https://www.space.com/15498-europa-sdcmp.html>

Researchers using the Hubble Space Telescope spotted a possible water plume jetting from Europa's south polar region in 2012. A different research team, after repeated attempts to confirm the observations, saw apparent plumes in 2014 and 2016. The researchers cautioned that the plumes haven't yet been fully confirmed, but they do provide a suggestion that there is water in Europa's ocean jetting to the surface.

Several spacecraft have conducted flybys of Europa (including Pioneers 10 and 11 and Voyagers 1 and 2 in the 1970s). The Galileo spacecraft carried out a long-term mission in the Jovian neighborhood between 1995 and 2003. Two missions are planned to visit Europa and other Jovian moons within the next decade, ESA's Jupiter Icy Moons Explorer (JUICE) mission and NASA's Europa Clipper. Pictures and data from the Galileo spacecraft suggest Europa is made of silicate rock, and has an iron core and rocky mantle, much like Earth does.

From fluctuations in Europa's magnetic field that suggests a conductor of some sort, scientists also think there is an ocean deep beneath the surface of the moon. This ocean could contain some form of life. This possibility of extraterrestrial life is one of the reasons interests in Europa remains high. In fact, recent studies have given new life to the theory that Europa can support life.

The surface of Europa is covered with cracks. Many believe these cracks are the result of tidal forces on the ocean beneath the surface. It's possible that, when Europa's orbit takes it close to Jupiter, the tide of the sea beneath the ice rises higher than normal. If this is so, the constant raising and lowering of the sea caused many of the cracks observed on the surface of the moon.

Obtaining samples of the ocean may not require drilling through the icy crust, should the repeated observations of possible plumes turn out to be actual jets of water. While researchers spotted evidence in 2012, 2014 and 2016, the true nature of the plumes — and why they appear sporadically — requires more observations.

The icy depths of the moons are thought to contain vents to the mantle much as oceans on Earth do. These vents could provide the necessary thermal environment to help life evolve. In 2016, a study suggested that Europa produces 10 times more oxygen than hydrogen, which is

similar to Earth. The moon may not need to rely on tidal heating to generate enough energy. Instead, chemical reactions would be enough to drive the cycle.

Exploration of Icy Ocean Worlds using Geophysical Approaches

<https://iopscience.iop.org/article/10.3847/PSJ/ac1272/pdf>

The text emphasizes the critical role of geophysical methods in advancing our exploration of icy ocean worlds, with a particular focus on the Galilean moon Europa and the Saturnian moons Titan and Enceladus. These celestial bodies are of substantial interest to the planetary science community owing to their potentially habitable subsurface oceans and active geological processes on their surfaces.

Despite the significant contributions from previous missions, such as the Galileo spacecraft, which provided valuable surface imaging of these moons, and the Cassini-Huygens mission that explored Saturn and its moons, much remains unknown regarding their internal structures and dynamics. The surfaces of these icy moons have been mapped extensively, yet understanding their interiors—the composition, dynamics, and characteristics of the ice shells and the underlying oceans—requires further investigation.

Geophysical techniques like geodesy, seismology, ice-penetrating radar, altimetry measurements, and electro-magnetometry are essential for acquiring detailed insights into the internal architecture of these icy worlds. These approaches can reveal crucial information about the thickness and dynamics of ice shells, the composition and behavior of subsurface oceans, and the interactions between deep interior materials and surface environments.

The exploration of these moons could answer a variety of pressing scientific questions. For example, understanding how material from the oceanic depths exchanges with the surface can provide context for assessing the habitability of these environments and any potential biosignatures that could indicate the presence of life.

The ongoing research into these icy ocean worlds aims to enhance our comprehension of their geophysical characteristics and astro biological potential. As data accumulation from geophysical observations increases, it will be pivotal in properly contextualizing the samples collected from their surfaces and interpreting signs that may suggest biological activity. Given the thickness of the ice shells, which can range from a few kilometers to tens of kilometers, direct sampling of the subsurface oceans proves challenging except in cases like Enceladus, where active plumes release ocean material onto the surface, providing a unique opportunity for study.

Through these detailed geophysical investigations, scientists hope to bridge the existing knowledge gap regarding these intriguing icy bodies and better understand the conditions that could support life beyond Earth.